

PROJECT REPORT

Import Data Using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-8263

1. INTRODUCTION

1.1 Project Overview

The project, Import Data Using Transform Maps (Spreadsheet), is a ServiceNow-based solution for centralizing employee information imported from spreadsheets. The solution uses an Import Set and staging table to receive spreadsheet records, a Transform Map to move the data into the Employee Test table, and Employee ID as the coalesce field to match existing records. The processed information is then presented through reports and an Employee Analytics Dashboard.

1.2 Purpose

The purpose is to reduce repetitive manual data entry, maintain structured employee records, prevent unnecessary duplicate records, support reporting by department and location, and provide controlled access to employee information through ServiceNow roles and ACLs.

2. IDEATION PHASE

2.1 Problem Statement

HR teams need a centralized ServiceNow Skill Wallet to manage employee skill and training information securely and consistently. The solution should support spreadsheet imports, accurate employee identification through Employee ID matching, structured employee information, analytics by department and location, and controlled access for authorized users.

2.2 Empathy Map Canvas

Area	Key Points
Says	Need employee information in one place; import spreadsheet data without entering every record manually; only authorized HR users should access or modify records.
Thinks	Manual entry can cause errors and duplicates; changed information should update the correct employee; reports should support department and location analysis.
Does	Collects employee information in spreadsheets, imports it into ServiceNow, reviews records, and uses reports/dashboard.
Feels	Concerned about inaccurate records, duplication, privacy and unauthorized access; more confident when imports complete successfully.
Goals	Centralize employee information, reduce manual effort, match employees using Employee ID, support reporting, and control access.

Pains / Challenges	Scattered spreadsheets, time-consuming manual updates, duplicate records, limited reporting visibility, and access-control requirements.
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2.3 Brainstorming

Idea Group	Brainstormed Ideas	Purpose / Benefit
Data Management	Central employee table; spreadsheet import; Import Set and Transform Map.	Reduce manual entry and keep data structured.
Record Matching	Use Employee ID as coalesce field.	Match existing employees and reduce duplicate records.
Information Enrichment	Maintain department and location information.	Provide useful context for reporting.
Analytics & Reporting	Employee list, location report, department report, dashboard.	Make employee information easier to analyse.
Security & Access Control	HR-specific roles and ACLs.	Limit access to authorized users.
Usability / Navigation	Expose relevant ServiceNow modules.	Make the solution easier to access.

Selected solution: a ServiceNow workflow that combines spreadsheet ingestion, employee matching, reporting, and role-based access control.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Experience	Interaction
Entice	HR identifies the need to centralize employee data.	Reviews employee information and prepares the spreadsheet.
Enter	HR starts the import process.	Uploads the spreadsheet through ServiceNow Import Sets.
Engage	Records are transformed into the target table.	Runs the Transform Map and uses Employee ID for matching.
Exit	Imported records are verified.	Checks employee name, email, department, Employee ID and location.
Extend	Employee information is used for reporting.	Views the Employee Analytics Dashboard by location and department.

3.2 Solution Requirement

FR No.	Functional Requirement
FR-1	Import employee information from a spreadsheet using ServiceNow Import Sets.
FR-2	Map spreadsheet columns to the Employee Test table using a Transform Map.
FR-3	Use Employee ID as the coalesce field for existing-record matching.
FR-4	Create new employee records and update matching records through transformation.
FR-5	Display employee name, email, department, Employee ID and location.
FR-6	Provide employee analytics by department and location.

NFR No.	Non-Functional Requirement
NFR-1	Usability – the import and record-management process should be simple for HR users.
NFR-2	Security – employee information should be controlled using ServiceNow roles and ACLs.
NFR-3	Reliability – valid spreadsheet records should be processed consistently.
NFR-4	Performance – bulk data processing should reduce manual record-by-record entry.
NFR-5	Availability – authorized users should be able to access records and dashboards.
NFR-6	Scalability – additional spreadsheet records can be processed using the same workflow.

3.3 Data Flow Diagram

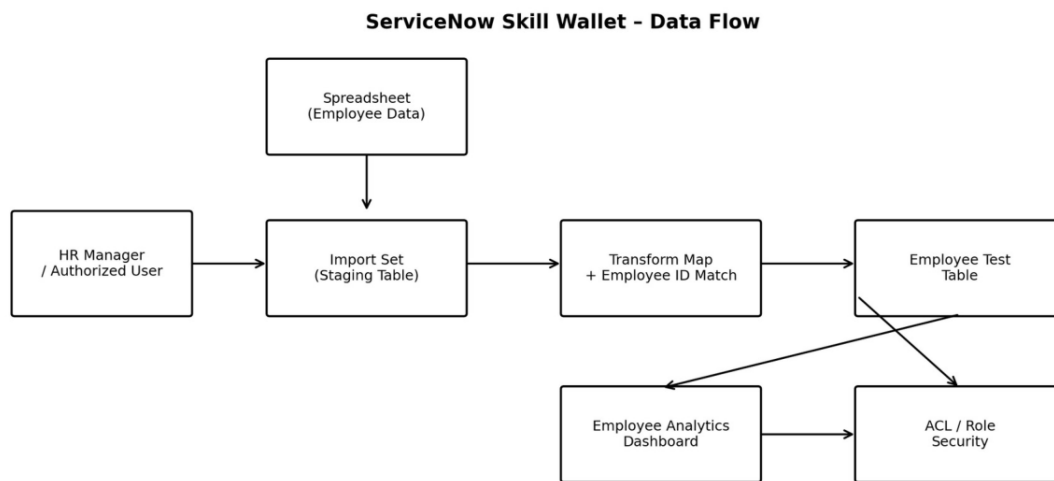


Figure 1: Data flow of the ServiceNow employee-data import solution.

Data flow: HR User / Authorized User → Spreadsheet → Import Set / Employee Import staging table → Transform Map → Employee Test table → Employee Analytics Dashboard. Security controls are applied through roles and ACLs.

3.4 Technology Stack

Layer / Component	Technology / Configuration
User Interface	ServiceNow Web UI
Import Layer	ServiceNow Import Sets and Employee Import staging table
Transformation Layer	Sample Spreadsheet Import Transform Map
Record Matching	Coalesce using Employee ID
Data Layer	Employee Test table / ServiceNow-managed platform database
Reporting	ServiceNow Reports and Employee Analytics Dashboard
Security	ServiceNow Roles and ACLs
Infrastructure	ServiceNow Cloud Platform
External APIs / ML	Not used in this project

4. PROJECT DESIGN

4.1 Problem Solution Fit

Area	Project-specific Content
Customer Segment(s)	HR personnel / HR managers maintaining employee information.
Problems	Import employee data accurately, avoid unnecessary duplicates, and make data available for reporting.
Triggers	Need for centralized employee data, reduced manual entry, record updates, and reporting.
Constraints	Consistent data structure, reliable Employee ID matching, and use within the existing ServiceNow environment.
Root Cause	Spreadsheet-based employee information requires manual handling and makes centralized reporting harder.
Solution	ServiceNow Import Set + staging table + Transform Map with Employee ID coalesce, followed by reports/dashboard.

4.2 Proposed Solution

The proposed solution is a ServiceNow-based employee data workflow. HR prepares the spreadsheet, loads it into the Employee Import staging table, and runs the Sample Spreadsheet Import Transform Map. Five source-to-target mappings are used: u_name → u_employee_name, u_email → u_email, u_department → u_department, u_employee_id → u_employee_id, and u_location → u_location. Employee ID is configured for coalesce so existing employees can be matched during re-import. The final records are displayed through reports and the Employee Analytics Dashboard.

4.3 Solution Architecture

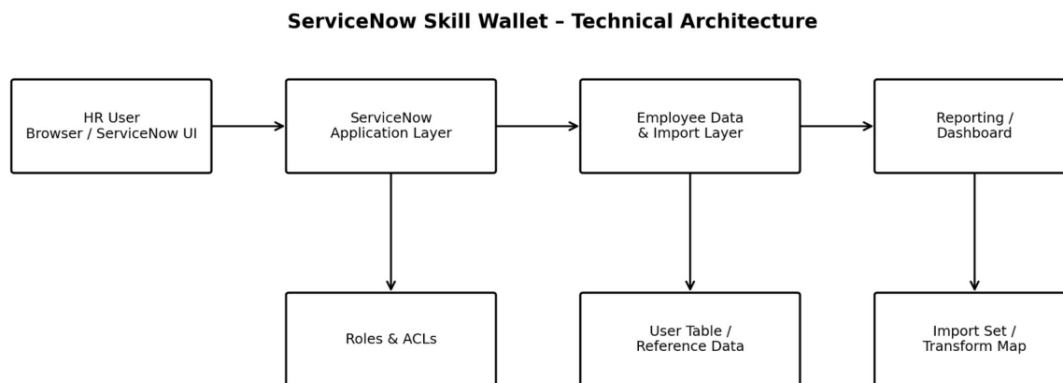


Figure 2: Technical architecture of the ServiceNow solution.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Scope	Story Points
Sprint 1	Data collection, Employee table, Import Set, Transform Map, transformation and validation.	12
Sprint 2	Coalesce/deduplication, new Excel data, reports and dashboard.	16
Total	Total completed story points.	28

Velocity = 28 story points / 2 sprints = 14 story points per sprint.

Final dataset: 17 employee records.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Scenario	Actual Result	Status
FT-01	Employee data field validation	Valid rows imported; missing name values were identified and corrected.	Pass
FT-02	Spreadsheet import	Rows loaded successfully into u_employee_import.	Pass
FT-03	Transform Map field mapping	All required source fields mapped to target fields.	Pass
FT-04	Coalesce duplicate prevention	Re-import test produced 4 updates and 0 inserts; no duplicate rows were created.	Pass
PT-01	Transform run-time test	Transform completed successfully in 0 seconds in the recorded test.	Pass
PT-02	Bulk import test	17 records processed with 0 errors.	Pass
PT-03	Identical re-import test	4 records were ignored on the recorded identical re-run; no errors.	Pass

User Acceptance Testing covered 8 test cases: 2 for data import and transformation, 2 for coalesce/deduplication, 3 for reports, and 1 for the dashboard. All 8 passed. Three identified bugs were closed: duplicate records on re-import, blank Employee Name values, and an incorrect Location chart type.

7. RESULTS

7.1 Output Screenshots

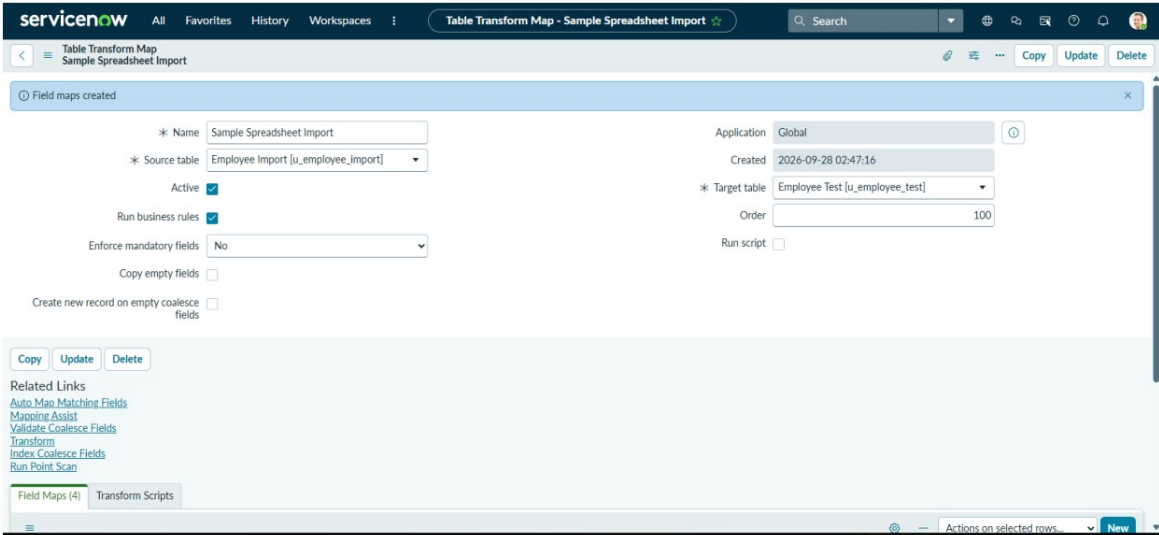


Figure 3: Sample Spreadsheet Import Transform Map configuration.

Field Maps (5)			
Transform Scripts			
Actions on selected rows... New			
	Source field	Target field	Coalesce
☐	u_name	u_employee_name	false
	u_email	u_email	false
	u_department	u_department	false
	u_employee_id	u_employee_id	false
	u_location	u_location	false
1 to 5 of 5			

Figure 4: Transform Map field mappings.

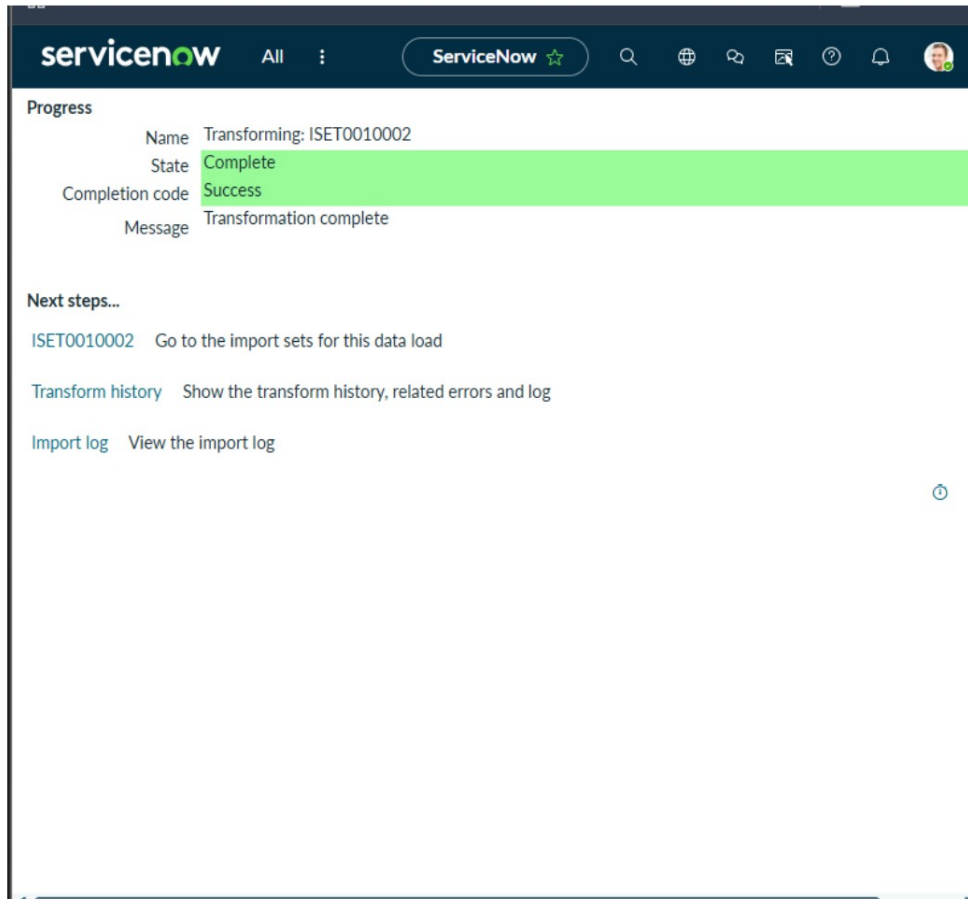


Figure 5: Successful transformation.

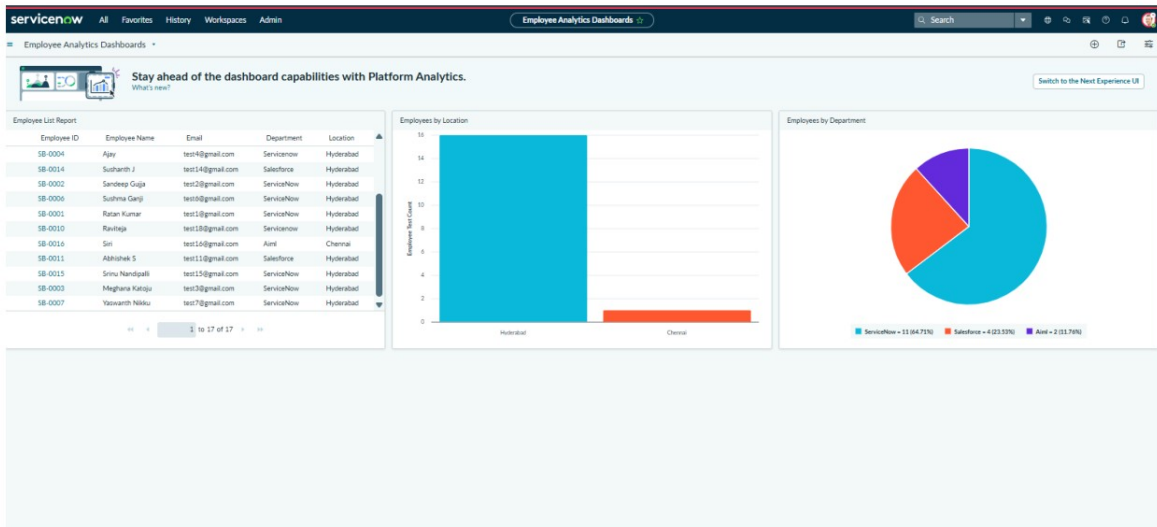


Figure 6: Employee Analytics Dashboard with list, location and department reports.

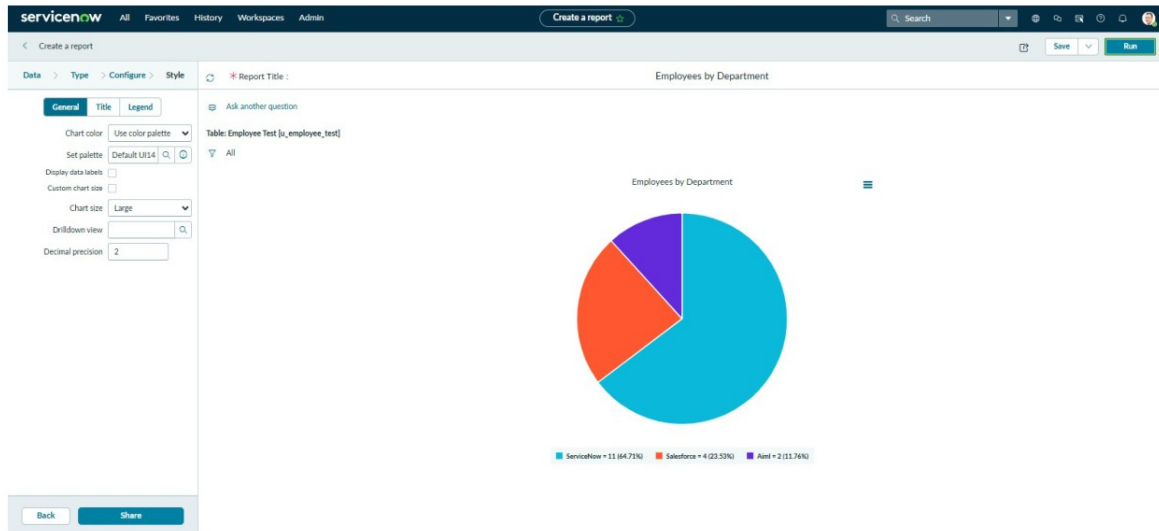


Figure 7: Employees by Department report.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Reduces repetitive manual data entry.	Depends on correctly structured spreadsheet input.
Employee ID coalesce supports reliable matching and duplicate prevention.	ServiceNow configuration and permissions are required.
Centralized employee records support reporting.	The current solution is focused on the defined employee dataset and fields.
Reports and dashboard provide quick visibility.	Changes to source columns may require mapping updates.
Roles and ACLs support controlled access.	No external public demo or standalone application is provided.

9. CONCLUSION

The project successfully demonstrates spreadsheet-based employee-data ingestion in ServiceNow using Import Sets and Transform Maps. The implementation stores structured employee records, uses Employee ID coalesce for matching, and provides reports and a dashboard for department and location analysis. Testing confirmed successful processing of the project dataset and closure of the identified defects.

10. FUTURE SCOPE

Future enhancements can include additional employee and training fields, automated scheduled imports, stronger validation rules for spreadsheet data, richer dashboard filters, audit-oriented reporting, notification workflows, and expanded role-based access for different HR responsibilities.

11. APPENDIX

Source Code: Not applicable as a standalone source-code repository; the project is implemented through ServiceNow configuration.

Dataset Link: Not applicable; the test dataset was supplied as a spreadsheet for the ServiceNow import process.

GitHub & Project Demo Link: Not applicable; the working demonstration is hosted in the ServiceNow Personal Developer Instance used for the project.

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